

Indices and surds - Set 2

TMUA practice questions by topic. Specification MM1.1, MM1.2, M2.7, M2.11, M2.12. 11 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Given that $\sqrt{a} - \sqrt{b} = 2$ and $a - b = 12$, find $\sqrt{a} + \sqrt{b}$.
- A** 4
B 8
C 10
D 6
E 24
F 3
- 2 If $x = 2^p$, $y = 2^q$ and $\frac{x^3}{y^2} = 2^5$ with $xy = 2^{10}$, find p .
- A** 3
B 7
C 8
D 4
E 6
F 5
- 3 Given that $(2 + \sqrt{3})^2 = a + b\sqrt{3}$ and $(2 + \sqrt{3})^{-1} = c + d\sqrt{3}$, where a, b, c, d are integers, find the value of $a + b + c + d$.
- A** 14
B 4
C 6
D 10
E 12
F 8

4 Which of the following is equal to $\frac{x^2\sqrt{x}}{\sqrt[3]{x}}$ for $x > 0$?

A x^3

B $x^{\frac{5}{2}}$

C $x^{\frac{7}{3}}$

D $x^{\frac{11}{6}}$

E $x^{\frac{17}{6}}$

F $x^{\frac{13}{6}}$

5 Find the value of $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{2+\sqrt{3}}$.

A 2

B $2 - \sqrt{3}$

C $\sqrt{3} + 1$

D 1

E $\sqrt{3}$

F $\sqrt{2}$

6 Given that $x = \sqrt{7} + \sqrt{3}$ and $y = \sqrt{7} - \sqrt{3}$, find $x^2 + y^2 - xy$.

A 24

B 16

C 4

D 20

E $20 - 2\sqrt{21}$

F 12

7 Which of the following statements is/are true for all positive real numbers a and b ?

I $\sqrt{a}\sqrt{b} = \sqrt{ab}$

II $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

III $\sqrt{a^2 + b^2} < a + b$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

8 The expression $\frac{3}{7-2\sqrt{3}} - \frac{3}{7+2\sqrt{3}}$ simplifies to

A $\frac{12\sqrt{3}}{6\sqrt{3}}$

B $\frac{42^7}{42}$

C $\frac{42}{37}$

D $\frac{0}{37}$

E $\frac{12\sqrt{3}}{37}$

F $\frac{12\sqrt{3}}{37}$

9 Which one of the following is the largest?

A $2\sqrt{4.5}$

B $\sqrt{2} + \sqrt{7}$

C $\sqrt{3} + \sqrt{6}$

D $\sqrt{4} + \sqrt{5}$

E $\sqrt{1} + \sqrt{8}$

10 a and b are positive real numbers. Which of the following statements is/are true?

I If $a^2 = b^2$ then $a = b$.

II If $2^a = 2^b$ then $a = b$.

III If $a^b = 1$ then $a = 1$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

11 For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?

A 9

B 1

C 10

D 0

E 2

F 100